Feature-Fusion Segmentation Network for Landslide Detection Using High-Resolution Remote Sensing Images and Digital Elevation Model Data

Xinran Liu, Yuexing Peng, *Member, IEEE*, Zili Lu, Wei Li, *Senior Member, IEEE*, Junchuan Yu, Daqing Ge, and Wei Xiang, *Senior Member, IEEE*

***Abstract*— Landslide is one of the most dangerous and frequently occurred natural disasters. The semantic segmentation technique is efficient for wide area landslide identification from high-resolution remote sensing images (HRSIs). However, considerable challenges exist because the effects of sediments, vegetation, and human activities over long periods of time make visually blurred old landslides very challenging to detect based upon HRSIs. Moreover, for terrain features like slopes, aspect and altitude variations cannot be sufficiently extracted from 2-D HRSIs but can be from digital elevation model (DEM) data. Then, a feature-fusion based semantic segmentation network (FFS-Net) is proposed, which can extract texture and shape features from 2-D HRSIs and terrain features from DEM data before fusing these two distinct types of features in a higher feature layer. To segment landslides from background, a multiscale channel attention module is purposely designed to balance the low-level fine information and high-level semantic features. In the decoder, transposed convolution layer replaces original mathematical bilinear interpolation to better restore image resolution via learnable convolutional kernels, and both dropout and batch normalization (BN) are introduced to prevent over-fitting and accelerate the network convergence. Experimental results are presented to validate that the proposed FFS-Net can greatly improve the segmentation accuracy of visually blurred old landslides. Compared to U-Net and DeepLabV3+, FFS-Net can improve the mean intersection over union (mIoU) metric from 0.508 and 0.624 to 0.67, the F1 metric from 0.254 and 0.516 to 0.596, and the pixel accuracy (PA) metric from 0.874 and 0.906 to 0.92, respectively. For the detection of visually distinct landslides, FFS-NET also offers comparable detection performance, and the segmentation is improved for visually distinct landslides with similar color and texture to surroundings.**

***Index Terms*— Digital elevation model (DEM), feature fusion, high-resolution remote sensing image (HRSI), landslide detection, semantic segmentation, Siamese network.**

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I. Introduction

LANDSLIDE is one of the most dangerous natural disasters and frequently occurs all over the world, which causes severe damages to human life and properties [1]. For example, the resurrection of an ancient landslide in the Oso community of Washington in 2014 resulted in 43 deaths, and a landslide occurred in Bijie County of China in 2017 claimed 27 lives and eight missing [2].

Landslides can be broadly categorized into old landslides and new ones. New landslides refer to those ones that are in the process of repeated landslide activities or have just stopped landslide activities and are still at risk of activities. Old landslides have a long history and remain inactive for a long period of time [3], which are temporary stable and may revive unexpectedly. For the task of landslide detection, both types require reliable identification.

Traditional landslide detection methods mainly rely on field investigation, aided visual interpretation of stereoscopic aerial photography, which is time consuming and resource-intensive and thus unsuited for wide-ranging applications. With the rapid advance of high-resolution remote sensing technology, abundant ground observation data are available, e.g., high-resolution remote sensing images (HRSIs), digital elevation model (DEM), and interferometric synthetic aperture radar (InSAR) data [4], [5], [6], which facilitate more efficient methods. Classic HRSI-based detection methods can be classified into two categories, i.e., pixel-based approaches and object-based image analysis (OBIA) methods. The former classify each pixel from spectral information [7], [8], [9], which lack geometric and contextual information and then are sensitive to salt-and-pepper noise [10], [11]. What is more, pixel-based approaches demand extensive parameter tuning, geometric correction, and coregistration of HRSIs [12]. By contrast, OBIA methods classify landslide areas by grouping objects with similar spectral, spatial, hierarchical, morphological, and structural characteristics [13], [14], [15], [16], [17]. However, they rely heavily on manual adjustment of key parameters such as the segmentation threshold and scale, which severely restricts its widespread application.

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Machine learning methods, including both shallow and deep network models, have attracted much research attention for detecting landslides in recent years. The former methods have been proposed for landslide detection, e.g., support vector machine, random forest, logistic regression, and many other classifiers [18], [19], [20]. But shallow network models have limited abstract presentation and self-learning capabilities and thus require expert intervention in challenging tasks such as landslide detection. Due to their superior abstraction and end-to-end learning capability, deep learning (DL) methods have been developed rapidly in the field of image processing. Convolutional neural network (CNN)-based object detection and semantic segmentation models have been widely employed in remote sensing image processing [21], [22].

Recently, many works have been proposed for landslide detection [23], [24], [25], [26], [27], [28], [29], [30], but most of them are focused on visually distinct landslides (most of them are new landslides), and few DL-based research works have been published on visually burred old landslide detection [30], [31]. Since DEM data can provide valuable terrain information for landslide detection, it is natural to extract semantic features from both HRSI and DEM and then fuse them for more reliable landslide detection [25], [26], which employ DEM data as an additional channel of input data and extract altitude information for recognition of visually distinct landslides. In this article, we develop a semantic segmentation model for more reliable recognition of visually burred old landslides, which extracts both rich details and semantic features from HRSI and DEM via a Siamese network and then fuses them in semantic space for deeper semantic feature distillation.

Our main contributions are summarized as follows.

1) The feature-fusion based semantic segmentation network (FFS-Net) model is developed for more reliable landslide detection. In the FFS-Net model, a Siamese network without weight sharing is designed to extract context, color and shape features from HRSIs and altitude change feature from DEM, which can extract rich detailed information via simple CNN. Then, the semantic feature extraction (SFE) module is designed to further distill deeper semantic features from the fused features for detecting visually blurred landslides. It is the first multisource heterogeneous data feature-fusion-based sematic segmentation work for visually blurred landslide detection based upon HRSIs and DEM data.
2) In the FFS-Net model, an enhanced SFE network is designed for better SFE based on the classic Deeplabv3+ network. From the Deeplabv3+ model, multiscale attention is introduced in the encoder to fuse the low- and high-level features from the backbone network to retain more detailed information, and transposed convolution layer replaces original mathematical bilinear interpolation in the decoder for better restoration of image resolution via learnable convolutional kernels. Moreover, dropout and batch normalization (BN) are introduced in the decoder to avoid over-fitting and accelerate the network convergence.
3) Sufficient performance evaluation experiments are carried out, and experimental results show that the proposed FFS-Net model achieves a great performance enhancement on visually blurred landslide detection.

II. Related Works

CNN-based DL methods are able to automatically extract semantic features from input images and classify them, and are widely employed in landslide detection recently [23], [24]. In [23], the CNN is combined with the regional growth algorithm (RSG_R) for new landslide detection. A Siamese CNN framework is proposed in [24] for new landslide detection, where a generative adversarial network (GAN) is employed to suppress the time-domain difference between dual-temporal HRSIs, and the Siamese network is utilized for pixel-level variation detection. However, it requires strict coregistration and geometric calibration for bitemporal images to suppress the time-domain difference. In [25], DEM data are exploited by the U-Net as an additional channel to HRSIs to provide terrain information like slope and aspect, and experimental results verify the performance enhancement by introducing DEM data for new landslide detection. In [26], an attention mechanism is introduced to a CNN-based model, and DEM data are utilized as an additional channel to improve the detection accuracy of landslides.

However, existing studies have focused mostly on the identification of visually distinct landslides (most of them are new landslides) [25], [26], [27], [28], [29], and few DL-based research works have been published on visually burred old landslide detection. Du et al. [30] tests several mainstream semantic segmentation models on the detection of old landslide with the best F1 score of 0.417. New landslides always present distinctly different visual features from surroundings, which facilitates the landslide detection. However, old landslides are much more difficult to recognize from HRSIs due to the fact that old landslides are often partially or totally covered by forest, or have been partly dismantled by erosion processes and human activities including agricultural and forest practices. Reliable recognition of visually blurred landslides remains a grave challenge [31]. Two examples are illustrated in Fig. 1, from which it is very difficult to recognize the old landslides from either HRSIs or DEM data. Specifically, we label the landslides and slopes in different colors in Fig. 2. As can be observed from Figs. 1 and 2, the landslides are so similar to the slopes that one cannot distinguish them. This calls for the utilization of more comprehensive information for old landslide recognition.

Based upon the fact that experts identify landslides from stereo images according to the slope, aspect, altitude change and other terrain information besides optical features, which is depicted in Fig. 3, it is reasonable to improve old landslide detection by smartly fusing both optical and terrain information, which is the main motivation of this work. We illustrate the visual and terrain features of landslides in Figs. 4–7. Specifically, the low-level features of landslides include visual features like texture features (the shadow-shaped boundary formed by projecting backwall/sidewall of landslides onto a 2-D HRSI, as shown in Fig. 4) and color features (especially for new landslides which differ greatly from surroundings in the color space, as shown in Fig. 5), and the high-level features

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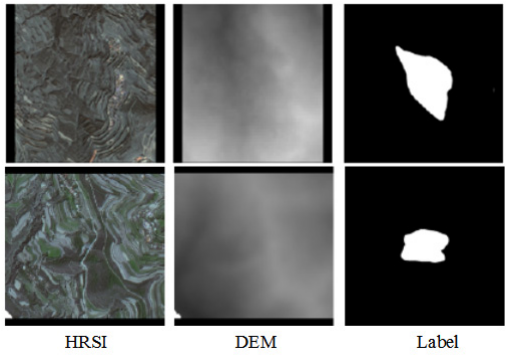


Fig. 1. Two old landslide samples in loess plateau dataset to show the difficulty to recognize old landslides from HRSI and DEM.

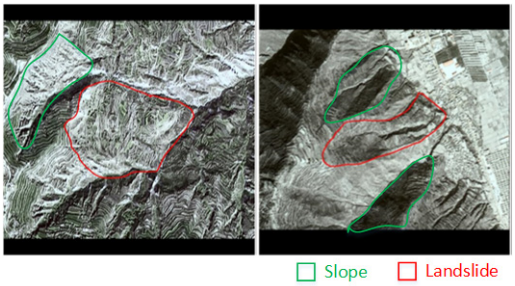


Fig. 2. Similarity of old landslides to slopes in loess plateau dataset.

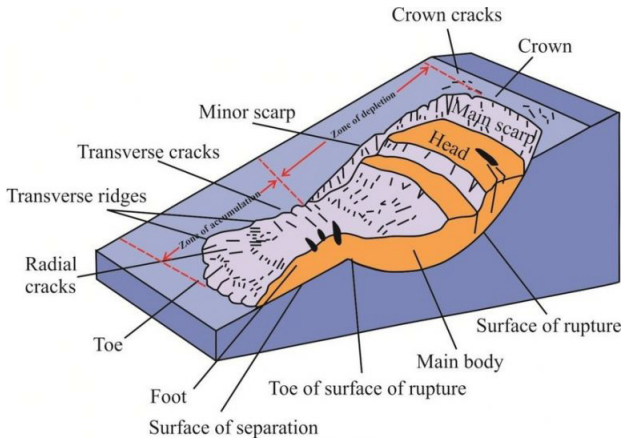


Fig. 3. Main components of the landslide [32].

of landslides refer to the abstract features like morphological characteristics (the tongue-like shape or horseshoe-like shape of the landslide accumulation areas in HRSIs as shown in Fig. 6, and densely distributed contour lines parallel to the back wall and sparsely distributed contour lines perpendicular to the side wall as shown in Fig. 7, where we transform the DEM into the contour map using the Gauss-Laplace edge operator). As can be observed from Figs. 4–7, the main scarp (also termed back wall), minor scarp (side wall) and accumulation body constitute the characteristic visual features in 2-D HRSIs in the form of texture and shape features, while the shapes, including the tongue-like or horseshoe-like shapes in accumulation area, the upper concavity and/or lower

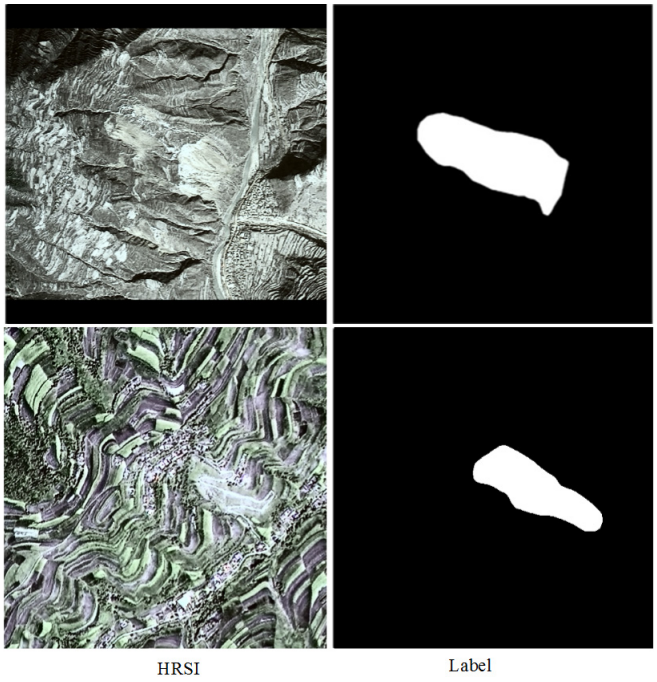


Fig. 4. Texture feature of old landslides, projected into 2-D HRSI from back walls and side walls.

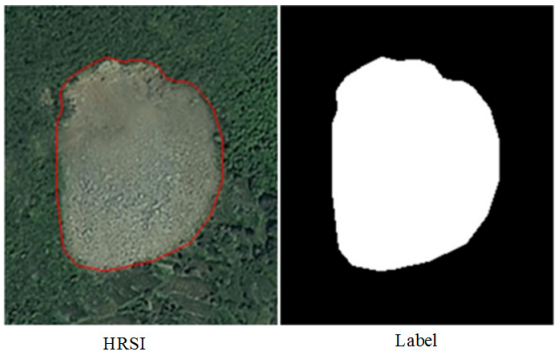


Fig. 5. Color feature of landslides, mainly the new landslides.

convexity on a slope, constitute the terrain features in the contour map in DEM data. Hence, in this work, we extract richer unique features from HRSIs and DEM data with the objective of improving the detection of landslides.

CNN-based semantic segmentation models extract high-level abstract semantic features and classify each pixel of an optical image such that the target objects are separated from the background and other inferring objects. The altitude information provided by DEM data is also employed in [25] and [26] as an additional channel of input data. To fully exploit the geological and geomorphological features from both HRSIs and DEM data, a feature-fusion landslide segmentation model, dubbed FFS-Net, is proposed in this article for detecting visually blurred landslides. The biggest difference between our model and existed models lies in the feature extraction and fusion methods. In our model, before fusing HRSIs and DEM data, feature transformation is independently performed on both the original HRSIs and DEM data via a Siamese network without weight sharing.

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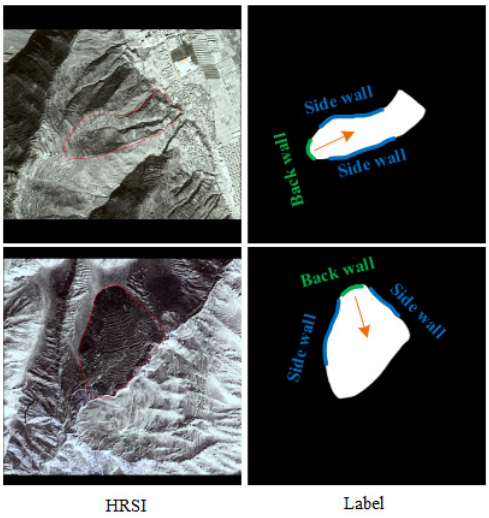


Fig. 6. Morphological characteristics in the shape of horseshoe or tongue of the deposits, where orange arrow denotes the sliding direction.

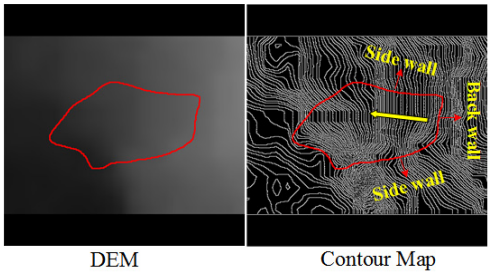


Fig. 7. Altitude change feature of landslides. Contour maps are constructed from DEMs via Gauss-Laplace edge operator, and the back walls are parallel to the contour lines while sidewalls are vertical to the contour lines. In contour map, yellow arrow denotes the sliding direction.

Compared with existed data fusion methods, the fusion of the extracted semantic features from two types of data is more consistent with the essence of semantic segmentation.

There are some existing works on feature extraction from heterogeneous data and feature fusion in the semantic space [33], [34], [35]. In [33], four schemes are proposed to fuse features extracted from the digital surface model (DSM) and aerial images, and comparative experiment results show that the early feature fusion scheme achieves the best semantic segmentation results. In [34], the DP-DCN model is proposed, which extracts spectral features and DSM's high-level features independently using two CNN branches, and then fuses features to feed to the decoder. In [35], similar to [34], region features are independently extracted from optical images and SAR images using two CNN branches, and the final result is obtained by the registration of two feature maps. Although the two CNN branches-based feature extraction and semantic feature fusion are similar, the proposed FFS-Net model differs greatly from the aforementioned models in that: 1) the SFE module is designed for deeper SFE from fused features, because the high similarity between landslides and surroundings demand deeper semantic features; 2) the squeeze-and-excitation (SE) and multiscale channel attention modules are introduced to combine the low-level and high-level features, which are of vital importance for old landslide detection; and 3) in the decoder, transposed convolution layers with learnable parameters replace the pure mathematical bilinear interpolation with result of better reconstruction resolution and less information redundancy, and both dropout layers and BN are introduced to avoid information redundancy and prevent over-fitting.

III. Proposed Method

As illustrated in Fig. 8, the proposed FFS-NET network model consists of three modules, i.e., the feature extraction and fusion module, the SFE module, and the decoder.

A. Feature Extraction and Fusion Module

The feature extraction module is a Siamese network, which consists of two identical CNNs without weight sharing, as shown in Fig. 9. The CNN is designed to be lightweight such that it can be a common feature extractor to be used easily in other networks. BN and rectified linear unit (ReLU) activation are employed for more stable gradient propagation and faster network training. Different from existing works, instead of regarding DEM and HRSIs as four-channel data, we extract color and morphological features from HRSIs and terrain features from DEM separately by the two branches of the Siamese network, and then fuse them at the higher feature level. The feature fusion process can be formulated as

$$T_{\text{fusion}} = \varphi_1(I_{\text{RGB}}) + \varphi_2(I_{\text{DEM}}) \tag{1}$$

where φ_1 and φ_2 represent the feature extraction functions performed independently by the two branches that do not share weights, I_{RGB} and I_{DEM} are the input heterogeneous data. $+$ denotes the pixelwise addition and T_{fusion} is the fused features. Before being fed to the Siamese network, the DEM data are translated into three-channel data to achieve data balance, better data expression, and data structure consistency with HRSIs for the sake of simplifying feature fusion model design.

B. Semantic Feature Extraction Module

ResNet101 [36] has been proven to have superior feature extraction capability, which is employed as the backbone network to distill semantic features for landslide detection. To extract and fuse both rich small-scale features like texture and steep elevation changes determined by scarps and large-scale features like the landslide shape, a multiscale attention module and an Atrous spatial pyramid pooling (ASPP) module are introduced to enhance landslide detection. The network structure is depicted in Fig. 8.

1) Backbone: Through the aforementioned Siamese network, visual features from HRSIs and terrain features from DEM are extracted, from which deep semantic features are further distilled. A pretrained ResNet101 network acting as the backbone is employed to extract features from T_{fusion}.

ResNet101 has four residual blocks, as shown in Fig. 8, where the structural sizes of convolution layers are listed in each block. For example, [3 $\times$ 3, 512] in block 4 denotes

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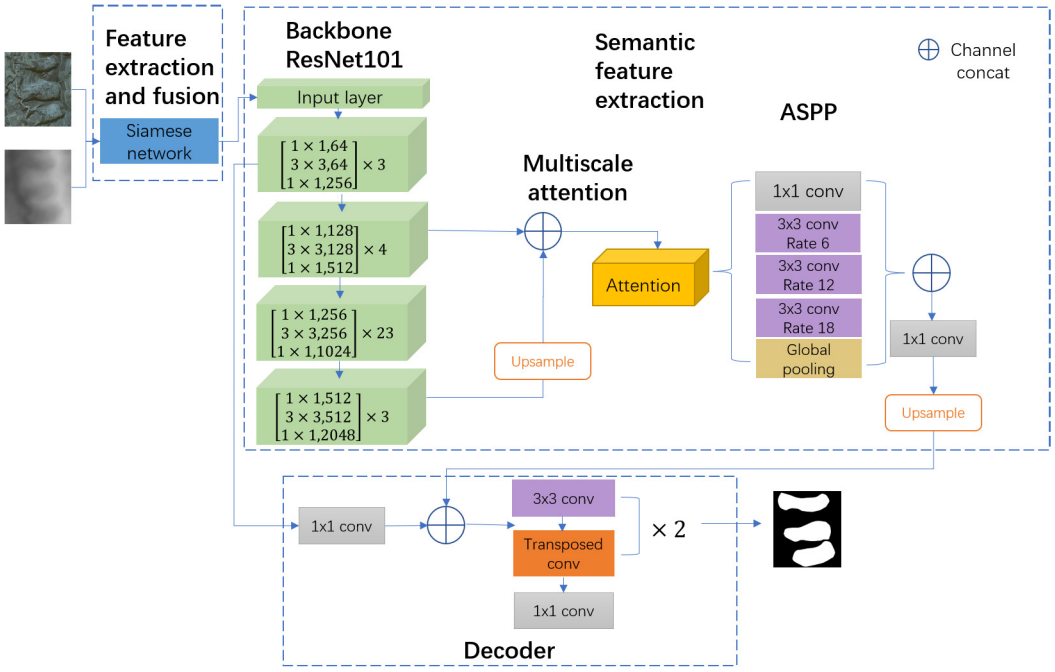


Fig. 8. Network structure of the FFS-Net model, including the feature extraction and fusion, SFE, and decoder modules, where [3 × 3, 512] denotes the size and number of the convolution kernels.

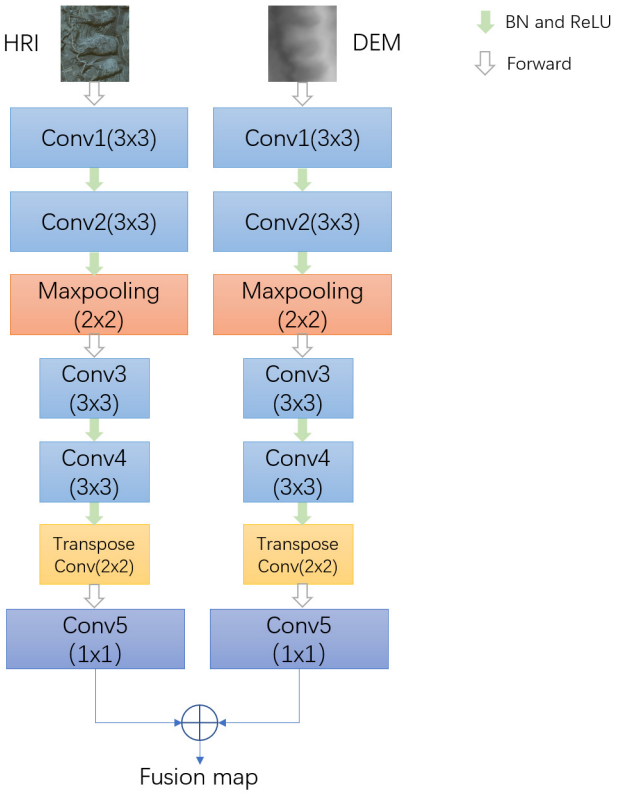


Fig. 9. Network structure of the feature extraction and fusion module, which is a Siamese network without weight sharing.

the convolution kernel size being 3 × 3 and the number of convolution kernels being 512. Then every block has a distinct resolution of feature maps. To retain more detailed features, after cross-testification, the output feature maps of block 2 are concatenated with those of block 4 after up-sampling before being fed the multiscale channel attention module for adaptive weighting multiscale features.

2) Multiscale Channel Attention Module: The multiscale channel attention module is designed to balance high-level semantic information for object classification and low-level detail information for pixel classification. Since successive convolution processing may incur large losses of detailed information, the low-level features yielded by block 2 are concatenated to the output of block 4 after up-sampling via channel fusion, and then automatically weighted by the channel attention module to achieve better sematic segmentation. The network structure is shown in Fig. 10. The channel attention module adopts the SE network [37] for weight learning between the channels of feature maps. As illustrated in Fig. 10, the input feature is first compressed from dimension $W \times H \times C$ to dimension $1 \times 1 \times C$ through the squeeze operation, where W, H and C are the width, height and channel of the feature map, respectively. After compression, the weight vector of dimension $1 \times 1 \times C$ for channel-level features over the global receptive field is learned by two fully connected layers, which perform the excitation operation and automatically learns the nonlinear relationship between channels. After the channel weight vector has been learned, the original input features are channel weighted by elementwise multiplication. The multiscale attention mechanism automatically weights low-level detail information from block 2 and high-level semantic information from block 4.

3) Atrous Spatial Pyramid Pooling : ASPP [38] is employed to further extract multiscale context information from the channelwise weighted features F', and the network structure is illustrated in Fig. 11. As discussed in Section I and depicted

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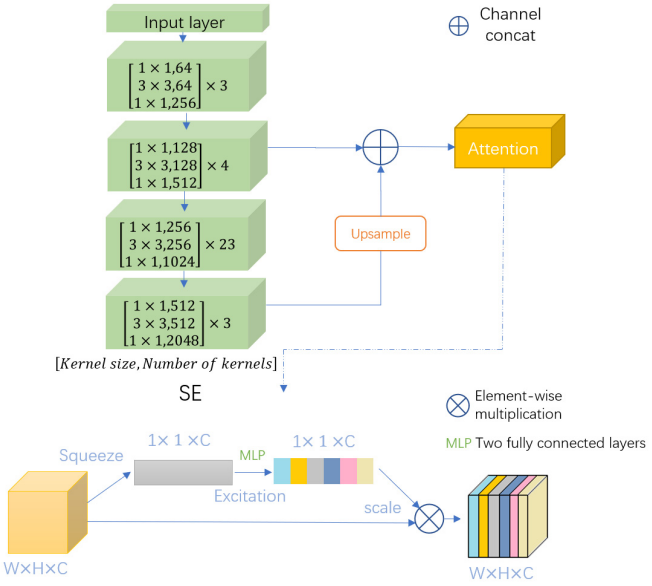


Fig. 10. Multiscale channel attention module with SE network.

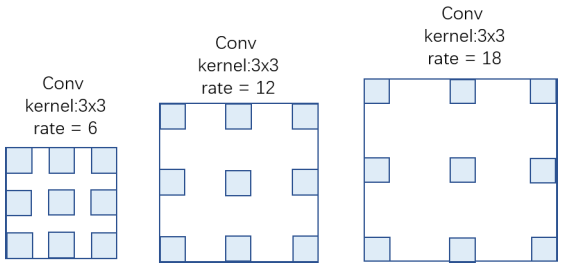


Fig. 11. Dilated convolution modules with different receptive fields via changing expansion rates [6, 12, 18].

in Fig. 10, the two main semantic features of landslides are small-scale textural features in HRSIs and contour patterns in DEM data driven by sudden drops in the elevation of major/minor scarps, the other one is the large-scale morphological features of landslides (main body and toe). Then a larger receptive field can help multiple-scale feature extraction, and hence ASPP is adopted here.

ASPP employs multiple parallel dilated convolutions with various expansion rates to extract features with different scales effectively, and then integrates the global context information through average pooling of image-level features. As depicted in Fig. 11, the dilated convolution module distills spatial features of different receptive fields through varying the expansion rate. The ASPP module outputs the high-level semantic feature map $F = \Phi(F')$.

C. Decoder

The decoder predicts and segments the landslide by taking both the low-level features yielded by the first block (B1) of the backbone network and the high-level semantic information yielded by the deep SFE module, as shown in Fig. 12.

In the decoder, transposed convolution layers reconstruct HRSI's resolution, which produce less information redundancy

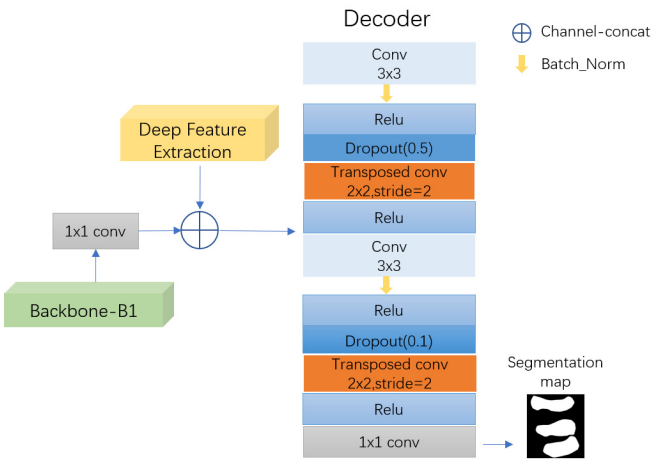


Fig. 12. Decoder network schematics.

and result in better feature mapping capability. Compared to the decoder in the Deepabv3+ model [39], which employs a pure mathematical bilinear interpolation without learnable parameters for up-sampling and thus leads to reduced learning capability, the transposed convolutional layers in our decoder can better restore the image resolution via learnable convolutional kernels through back propagation.

In the designed decoder, dropout layers are used to avoid information redundancy and prevent over-fitting, BN layers are used to constrain the data distribution so as to avoid both gradient explosion and gradient disappearance and to aid in network convergence acceleration, and ReLU is adopted as the activation function to accelerate the training speed, increase network sparsity, and avoid gradient disappearance.

IV. Datasets and Settings

In this section, the proposed FFS-Net model is evaluated and compared with several state-of-the-art reference models. All experiments are conducted on a server whose configuration parameters are as follows: Intel(R) Xeon(R) Platinum 8163 @ 2.50 GHz, eight cores, 32-GB RAM; Tesla V100-SXM2 16G; PyTorch 1.2.0.

A. Data Preprocessing

Two datasets are employed for performance evaluation, one is Loess Plateau dataset which consists of visually blurred old landslides, the other is Bijie dataset which consists of both old and new landslides.

1) Loess Plateau Dataset: The study area locates in Loess Plateau of China, where the soil parent material is eluvial and accumulative parent material. The soil is generally viscous, with weak water permeability, corrosion resistance and erosion resistance, which is directly related to the occurrence of landslide. Soil erosion caused by sparse vegetation and concentrated precipitation is another reason for landslide formation.

The surface features of old landslides such as color and texture are similar to surroundings. To make matters worse, the main bodies of some old landslides are transformed into farmland or residential area (see samples in Fig. 9). These kinds of

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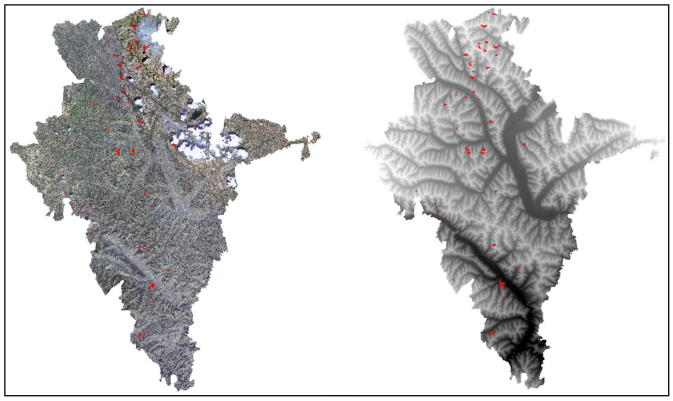


Fig. 13. Landslide distribution in county A, where red dots denote landslides.

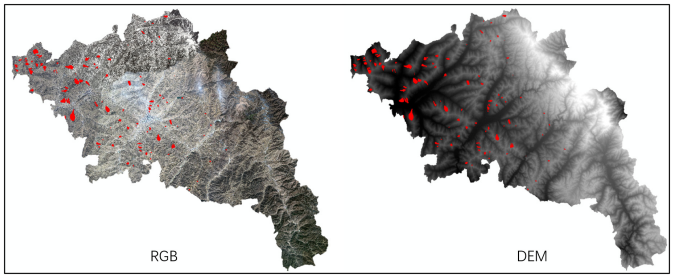


Fig. 14. Landslide distribution in county B, where red dots denote landslides.

changes on surface features make it very difficult to reliably detect landslides.

Old landslide samples were measured in 2018 by the Gaofen-1 satellite, and the dataset were made by the Institute of Remote Sensing and Digital Earth (RADI) under the Chinese Academy of Sciences (CAS). There are 195 old landslides in Loess Plateau dataset, which are split into the training, validation, and test sets with a ratio of 6:2:2. The resolution of HRSIs is 2 m/pixel, and the original resolution of DEM data is 30 m/pixel, which are interpolated to match the resolution of the HRSIs.

The distribution maps of the old landslides in two Chinese counties are shown in Figs. 13 and 14. Clearly, old landslides are widely distributed.

2) Bijie Dataset: The study area is located in Bijie County, Guizhou Province, China. The dataset contains both new and old landslides. Ji's team has made this dataset open for the benefits of fellow researchers, including both HRSIs with a resolution of 0.8 m/pixel and DEM data with a resolution of 2 m/pixel [26]. The DEM data are interpolated to match the higher resolution of the HRSIs. This dataset contains 770 landslide's images captured by the Triplesat satellite from May to August 2018.

As shown in Fig. 15, compared to Loess Plateau dataset, landslides in the Bijie dataset have much more visually distinct features different from surroundings in terms of color and shape, which facilitate the semantic segmentation of landslides. The samples in Loess Plateau dataset are visually blurred old landslides, which have nearly invisible difference from surroundings in HRSIs.

TABLE I
DATASETS PARTITION

	Training	Validation	Test
Bijie dataset	2772	154	154
Loess Plateau dataset	600	31	32

3) Data Preprocessing: Landslides' sizes vary considerably, and it is necessary to resize the samples to the same dimension so as to facilitate model training. To keep the same aspect ratio, we crop the images to 256×256 for Bijie dataset and 512×512 for Loess Plateau dataset by scaling and zero padding. Histogram equalization is implemented as well as data augmentation for the training set. The dataset is partitioned into the training, validation and test sets with a ratio of 6:2:2, and the sample numbers are listed in Table I.

B. Performance Metrics

The following five performance metrics are employed, i.e., pixel accuracy (PA), precision, recall, F1-score and mean intersection over union (mIoU), which are defined as follows:

$$\mathrm{PA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{2}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{3}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{4}$$

$$\mathrm{F1} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}} \tag{5}$$

$$\mathrm{mIoU} = \frac{1}{2} \times \left(\frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} + \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FN}+\mathrm{FP}} \right) \tag{6}$$

where TP, TN, FP, and FN are landslide pixels accurately predicted, nonlandslide pixels accurately predicted, and landslide pixels mistaken for nonlandslide, nonlandslide pixels mistaken for landslide, respectively.

C. Reference Models

Although landslide detection model [26] and heterogeneous information fusion models [33], [34], [35] have been published, no paper has released the source codes of their models, so it is impossible for us to reproduce their results for comparative purposes. Moreover, we cannot compare our model directly with heterogeneous information fusion models [33], [34], [35] due to the following three reasons: 1) the targets of semantic segmentation models are not landslide; 2) aerial images or DSM or SAR are employed, which are much different from HRSIs and DEM data; and 3) no source codes are released such that there is no way we are able to reproduce the same results presented in their papers. Therefore, we had no choice but to validate the advantages of our model via ablation experiments.

To benchmark the performance of the proposed FFS-NET model and validate the effectiveness of heterogeneous information fusion, two mainstream semantic segmentation models, i.e., U-Net [40] and Deeplabv3+ [39], are also evaluated on

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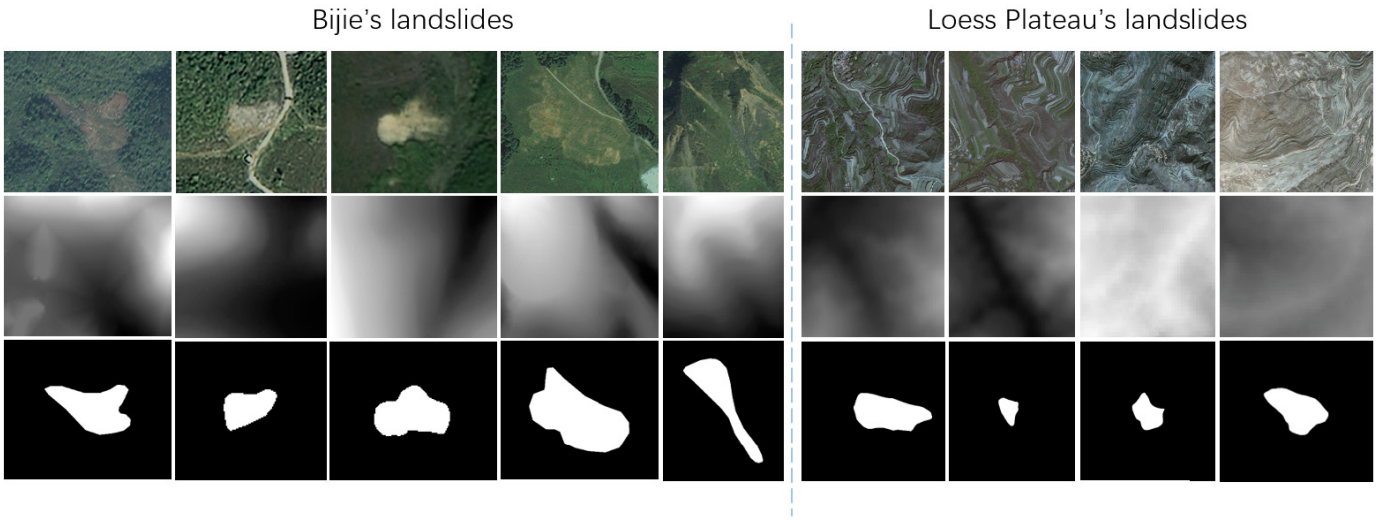


Fig. 15. Landslide samples of the two study areas. (Top to bottom) Image, DEM, and label.

the same datasets. Moreover, three distinct schemes are tested for all the three comparison models.

1) Scheme 1, where only HRSIs are used and this scheme is denoted by *Only-image* thereafter.
2) Scheme 2, where both the HRSIs and DEM data are used, and the DEM data are fused with the HRSIs in data layer to form the forth channel data in addition to the RGB channels of the HRSI. This scheme is denoted by *Channel-fusion*.
3) Scheme 3, where both the HRSIs and DEM data are used, and features distilled from the DEM and HRSIs are fused in a higher feature layer. This scheme is the developed in this work and denoted by *Feature-fusion* thereafter.

For all the comparison models, stochastic gradient descent is chosen as the optimizer with learning rate 0.007, momentum 0.9 and weight decay 0.0005. The cross entropy is adopted as the supervised loss function, where landslide pixels are labeled 1 and others 0

$$\text{Loss} = -y_i \log(\hat{y}_i) + (1 - y_i)\ \log(1 - \hat{y}_i) \tag{7}$$

where y_i and $\hat{y}_i$ denote the ground truth and the predicted one, respectively.

V. Results and Discussion

In this section, we present both experimental results and detailed analysis with visualization.

A. Overall Experiments

Three models with three types of data utilization schemes are evaluated on the both datasets.

1) Experimental Results on Loess Plateau Dataset: We list the numerical results in Table II, from which one can observe the followings.

1) The proposed FFS-Net model with the *Feature-fusion* scheme achieves the best performance among all the comparison models. Specifically, when compared to the original U-Net and Deeplabv3+ models, the F1 score increases from 0.254 and 0.516 to 0.596, the mIoU metric improves from 0.508 and 0.624 to 0.67, and the PA is up from 0.874 and 0.906 to 0.92. These improvements clearly demonstrate the efficacy of the proposed model for old landslide detection.
2) The DEM data can provide useful information for old landslide detection in the proposed FFS-Net model. It is evident that both fusion schemes for the FFS-Net model outperform their *Only-image* scheme. For example, the mIoU gains of the *Channel-fusion* and *Feature-fusion* schemes are 0.5% and 3.8% over the *Only-image* scheme. This performance gain is also achieved by the Deeplabv3+ model. As for the U-Net model, the backbone network is not effective enough to extract sufficient features for old landslide detection, and thus both DEM fusion schemes can contribute little.
3) *Feature-fusion* is more efficient to distill semantic features for old landslide detection than *Channel-fusion* on the small dataset. When the old landslide dataset has sufficient volume and variety, the *Channel-fusion* scheme performs no worse than its *Feature-fusion* counterpart.

TABLE II
Quantitative Experimental Results on Loess Plateau Dataset

Method	Precision	Recall	F1	mIoU	PA
U-Net					
Only image	0.384	0.190	0.254	0.508	0.874
Channel-fusion	0.466	0.137	0.212	0.501	0.885
Feature-fusion	0.428	0.185	0.258	0.513	0.880
Deeplabv3+					
Only image	0.615	0.444	0.516	0.624	0.906
Channel-fusion	0.688	0.418	0.520	0.630	0.913
Feature-fusion	0.632	0.467	0.537	0.636	0.909
FFS-Net					
Only image	0.703	0.416	0.523	0.632	0.914
Channel-fusion	0.655	0.457	0.538	0.637	0.911
Feature-fusion	0.699	0.519	0.596	0.670	0.920

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For small datasets in practice, the Siamese network designed in this work can efficiently extract semantic features and keep a less complicated network such that the model can be trained to converge on the small dataset. This is validated by the remarkable performance gains achieved by the *Feature-fusion* scheme over the *Channel-fusion* scheme in the FFS-Net model, where the mIoU gain is 3.3% and the F1 gain is 5.8%.

4) The U-Net model presents the worst performance because the feature extraction network is too simple to effectively distill semantic features for old landslide detection. The Deeplabv3+ model greatly enhances the network capacity by the use of dilated convolution and ASPP on top of ResNet101, and thus achieves the state-of-the-art performance in semantic segmentation. The proposed FFS-Net model stems from Deeplabv3+ and is specially optimized by feature fusion and attention mechanism in consideration of the characteristics of visually blurred old landslides.

The graphical results are presented in Fig. 16. As can be seen from Fig. 16, the old landslides segmented by the proposed FFS-Net have more accurate shapes than all other reference models, which is consistent with the numerical results.

2) Experimental Results on Bijie Dataset: The quantitative experimental results on Bijie's landslide detection are listed in Table III, while the graphical results are presented in Fig. 17. As can be seen from Table III and Fig. 17, the following observations can be made.

1) All the comparison models perform much better than that on the Loess Plateau's landslide. As aforementioned, Bijie's landslides possess easily recognized visual features like remarkable brightness and special shapes. Even the worst-performing U-Net achieves 0.838 in mIoU, which is 25% higher than the best performance achieved on Loess Plateau's landslide detection by the proposed FFS-Net. In this sense, all the models are effective in detecting visually distinct landslides.
2) Thanks to the easily recognized visual features, it is not necessary to fuse the features distilled from the DEM data for the DeepLabV3+ and FFS-Net models which have great feature distillation capability. This can be deduced from the results that the *Only image* scheme performs better than the two fusion schemes. Moreover, the *Feature-fusion* scheme performs worse than its *Channel-fusion* counterpart, which further verifies the feature extracted from DEM degrades the performance of visually distinct landslide detection. But this is not the case for the U-Net model, where the *Feature-fusion* scheme performs the best and the *Only image* scheme has the worst performance. The reason is that the U-Net model is not capable enough to extract sufficient features for landslide detection, which is consistent with the analysis in the case old landslide detection on the Loess Plateau dataset.
3) Although the proposed model performs slightly worse than the best one, it still offers a performance good enough to detect those visually distinct landslides on the Bijie dataset. As shown in Fig. 17, FFS-NET can also realize accurate segmentation on the portions of landslides with pronounced features, e.g., the case in Fig. 17(a)–(c). When there are interfering objects with similar color and texture to the landslides, like the cases in Fig. 17(d)–(f), the FFS-Net model with the *Feature-fusion* scheme has a better landslide segmentation effect with more accurate boundary, which are highlighted by the red circles. Then it is reasonable to conclude that the DEM data can help segment the landslide edge by excluding nonlandslide interfering objects with similar color and texture, and then reduce the misjudged areas.

TABLE III
QUANTITATIVE PERFORMANCE RESULTS ON THE BIJIE DATASET

Method	Precision	Recall	F1	mIoU	PA
U-Net					
Only image	0.816	0.841	0.828	0.838	0.972
Channel-fusion	0.852	0.809	0.830	0.840	0.973
Feature-fusion	0.851	0.846	0.848	0.855	0.975
Deeplabv3+					
Only image	0.894	0.888	0.891	0.892	0.982
Channel-fusion	0.878	0.882	0.880	0.882	0.980
Feature-fusion	0.889	0.849	0.869	0.873	0.979
FFS-Net					
Only image	0.895	0.877	0.886	0.888	0.982
Channel-fusion	0.888	0.867	0.878	0.880	0.980
Feature-fusion	0.876	0.857	0.867	0.871	0.979

B. Ablation Experiments

To further analyze the performance of the proposed model, we carry out ablation experiments on the two datasets.

1) Ablation Experiments on Loess Plateau Dataset: The experimental results are shown in Table IV, where *BASE* denotes the Deeplabv3+ model as a baseline, *Feature-fusion* denotes the proposed Siamese network-based feature extraction and fusion module, *Multiscale* denotes the proposed multiscale channel attention module, and *Improved decoder* denotes the modification of decoder by replacing bilinear interpolation with transposed convolution layers and the introduction of both dropout layers and BN layers. The results show that the following can be obtained.

1) When only HRSIs are utilized for old landslide detection, the modifications on the encoder and decoder of the baseline Deeplabv3+ model are effectiveness, and the F1 and mIoU performance are increased from 0.516/0.624 to 0.522/0.631 and 0.523/0.32. The overall improvement is significant, i.e., F1 score reaches 0.530 and the mIoU arrives at 0.634. From the experimental results, we can safely say that the modification of the Deeplabv3+ is effective.
2) When both HRSIs and DEM data are employed, the proposed FFS-Net model can greatly enhance the recognition performance. Specifically, the Feature fusion and the modification of Deeplabv3+ can improve the F1 score from 0.516 to 0.596, and the mIoU metric is significantly improved from 0.624 to 0.670.

As can be seen from the above ablation experimental results, it can be concluded that the high similarity of surroundings

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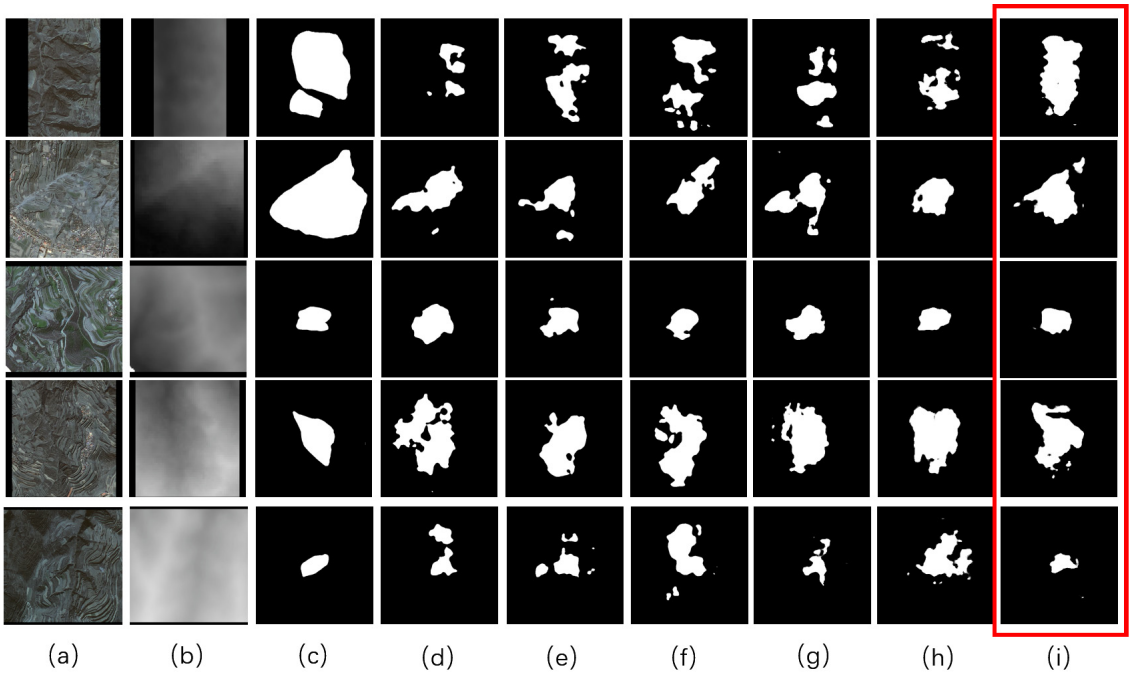


Fig. 16. Graphical experiment results on loess plateau's landslides. (a)–(c) RGB image, DEM, and label. (d)–(f) Deeplabv3+: RGB image, RGB image + DEM (four-channel data), and RGB + DEM with feature fusion. (g)–(i) Ours: RGB image, RGB image + DEM (four-channel data), and RGB image + DEM with feature fusion.

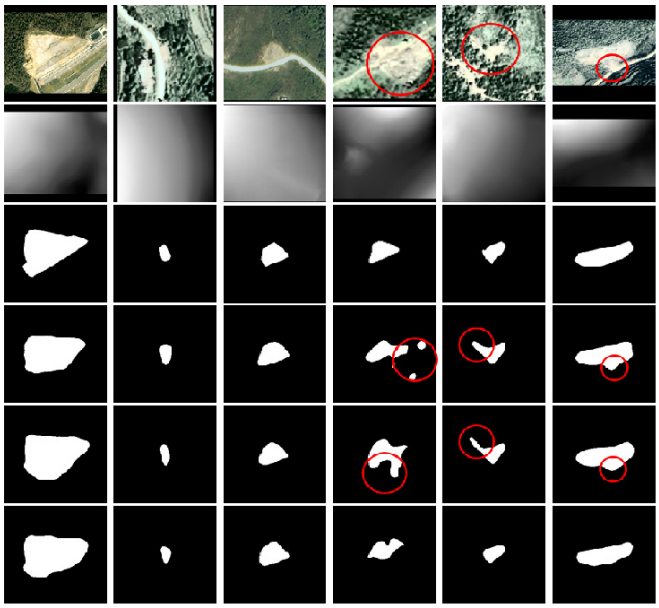


Fig. 17. Graphical detection results of Bijie dataset, where nonlandslide areas with similar characteristics around the landslides are marked in the figures. (Top to bottom) RGB image, DEM, label, using only RGB image, RGB image + DEM (four-channel data), and RGB image + DEM with feature fusion.

to especially old landslides, poses the greatest challenge to semantic segmentation model, and the visual features from the HRSIs are insufficient to reliably detect landslides. The proposed FFS-Net model can greatly improve the detection performance by extracting useful features from the DEM data and then fusing them at higher layers with the visual semantic features from HRSIs.

2) Ablation Experiments on Bijie Dataset: The experimental results are listed in Table V. However, we observe the opposite results for visually distinct landslides on the Bijie dataset. The baseline model achieves the best performance, and a worse performance is observed with the introduction of more modules. Specifically, the mIoU score reaches its peak value of 0.892 given by the baseline model without the use of the DEM data. When the DEM data are used, the introduction of *Feature-Fusion* or *Multiscale* will reduce the mIoU score by 1.9% and 0.4%, respectively. The same trends are observed for the F1 score.

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TABLE IV
ABLATION EXPERIMENTS ON LOESS PLATEAU DATASET

	Precision	Recall	F1	mIoU	PA
B: Base model, i.e., Deeplabv3+ M: Multi-scale attention I: Improved decoder F: Feature fusion					
B	0.444	0.615	0.516	0.624	0.906
B+I	0.424	0.679	0.522	0.631	0.913
B+M	0.408	0.678	0.523	0.632	0.911
B+I+M	0.433	0.683	0.530	0.634	0.913
B+F	0.467	0.632	0.537	0.636	0.909
B+F+I	0.441	0.692	0.538	0.639	0.914
B+F+M	0.493	0.653	0.562	0.649	0.913
B+F+M+I	0.699	0.519	0.596	0.670	0.920

TABLE V
ABLATION EXPERIMENTS ON BIJIE DATASET

	F1	mIoU
B	0.891	0.892
B + M	0.886	0.888
B + F	0.869	0.873
B + F + M	0.867	0.871

As for visually distinct landslides, easily recognized visual features are conducive to reliable detection. For the excellent visual feature extractor such as the ResNet101-based Deeplabv3+ model, there is no need to introduce additional modules to extract elevation feature from the DEM data, because the enhancement on segmentation performance may not compensate for the increased model complexity.

3) Convergence Analysis: The training loss curves and validation loss curves are also presented for performance comparison. As shown in Fig. 18, FFS-Net offers better generalization ability and effectiveness on Loess Plateau dataset. Specifically, the Deeplabv3+ model exhibits an evident overfitting phenomenon because the validation loss curve increases with the iteration, indicating a poor generalization ability. On the other hand, the validation loss curve of our proposed FFS-Net gradually decreases, indicating the improved generalization ability. Similar results are observed on Bijie dataset.

4) Complexity Analysis: We also compare the complexity of models, and the results on parameters and GFLOPs are listed in Table VI, where FLOPs stands for a billion floating point operations per second. Moreover, we analyze the time complexity of models in terms of epochs and execution time. The input HRSIs are tensors with size of (3, 256, 256), and BASE model is Deeplabv3+. The last model is the proposed FFS-Net with feature fusion scheme, whose inputs include HRSIs and DEM data. Compared to the baseline model, the proposed FFS-Net model pays the increase on acceptable training parameters, calculation, and execution time, but achieves great improvement on old landslide detection, i.e., F1 score increases from 0.516 to 0.596 (15.5%) and mIoU increases from 0.624 to 0.670 (7.4%).

5) Visualization Results: Moreover, we visualize the results of the *Feature-fusion* scheme and *Only-image* scheme of FFS-Net model on Loess Plateau dataset, where gradient-weighted class activation mapping (Grad-CAM) visualization method [41] is employed. As a popular visualization tool, Grad-CAM has a strong class distinguishing ability, and can find the most relevant features of the discriminating class through the gradient return of the target class. We visualize the middle layer of the decoder of our model, and illustrate some visualization results in Fig. 19. Clearly, the *Feature-fusion* scheme achieves a more accurate landslide boundary, from which the main scarp, minor scarp, and leading edge features of the landslide are detected more accurately. It is further verified that the introduction of the DEM data through feature fusion provides an additional valid source of information for landslide identification.

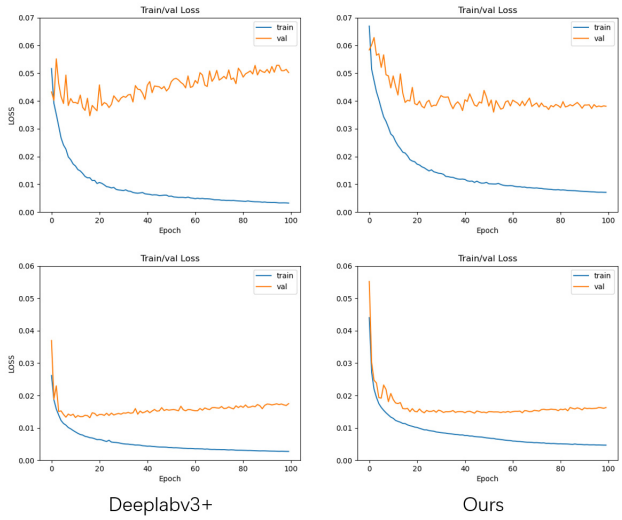


Fig. 18. Loss curves comparison of FFS-Net and Deeplabv3+ on two datasets. (Top to bottom) Loess Plateau dataset and Bijie dataset.

TABLE VI
TRAINABLE PARAMETERS AND COMPLEXITY OF MODELS

	Params(M)	GFLOPs	Epochs	Time (sec)
B	59.34	22.19	19	2090
B + F	59.93	34.67	48	7344
B + M	63.96	37.49	39	6786
B + F + M	64.55	49.96	52	9672

C. Cross Validation

To optimize the proposed FFT-Net model, four comparison experiments are carried out, including: 1) scheme of backbone network; 2) fashion of semantic feature fusion in the SFE module; 3) feature extraction scheme of the Siamese network; and 4) feature fusion scheme in the decoder.

1) Scheme of Backbone Network: Three widely used backbone networks, including the ResNet 101, ResNet 50, and Xception, are tested on the Loess Plateau dataset, and the experimental results are given in Table VII. As can be observed from Table VII, the ResNet101 scheme achieves the best F1 and mIoU scores. As a result, the ResNet 101 is chosen as the backbone network in the proposed FFS-Net model.

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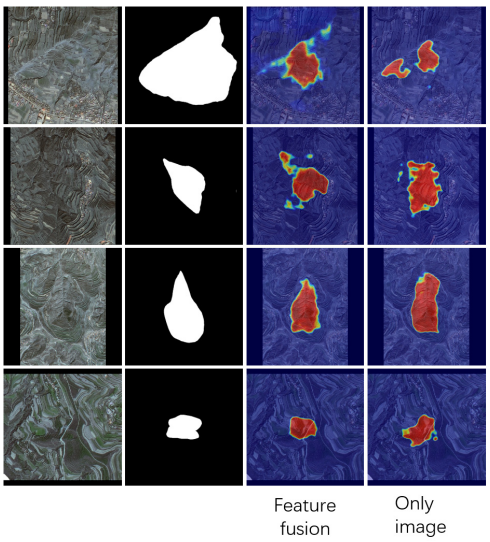


Fig. 19. Visualization results by Grad-CAM on loess plateau's landslides.

TABLE VII
COMPARATIVE EXPERIMENT ON DIFFERENT BACKBONE NETWORKS

Backbone	Precision	Recall	F1	mIoU	PA
ResNet101	0.699	0.519	0.596	0.670	0.920
ResNet50	0.652	0.488	0.566	0.652	0.914
Xception	0.581	0.443	0.502	0.616	0.901

TABLE VIII
EXPERIMENTAL RESULTS ON FUSION FASHIONS OF THE PROPOSED MODEL

Method	Precision	Recall	F1	mIoU	PA
Early Fusion	0.699	0.519	0.596	0.670	0.920
Late Fusion	0.624	0.402	0.489	0.612	0.905

2) Fashion of Semantic Feature Fusion: Semantic features can be fused before or after the deeper SFE, that is, it can be inserted before the backbone Resnet101 (early fusion) or the decoder (late fusion). The comparative results on the Loess Plateau dataset are listed in Table VIII, from which one can see that early feature fusion is more conducive to landslide detection. This observation is consistent with the comparison results in [33], which validates that early fusion in the encoder outperforms the other three fashions of feature fusion, i.e., late fusion in the encoder, fusion in the decoder, and fusion in both the encoder and decoder.

3) Feature Extraction Scheme of the Siamese Network: The Siamese network is designed for SFE from HRSIs and DEM data to present more comprehensive information for reliable recognition of landslides. Weight sharing is a usually deployed scheme for Siamese network, which can reduce the parameter number. However, in this article, the Siamese network extracts different semantic features of landslides from the heterogeneous HRSI and DEM data, and it is reasonable that the two branches of the Siamese network extract features independently. To testify this thought, comparative experiments are carried out, and the experimental results are listed in Table IX, which verify the independent feature extraction method via no weight sharing scheme behaving better than the weight sharing scheme.

TABLE IX
COMPARATIVE RESULTS OF WEIGHT SHARING SCHEME OF THE SIAMESE NETWORK

	Precision	Recall	F1	mIoU	PA
No weight sharing	0.699	0.519	0.596	0.670	0.920
Weight sharing	0.618	0.513	0.561	0.647	0.909

TABLE X
COMPARATIVE RESULTS OF FEATURE FUSION SCHEME IN THE DECODER

	Precision	Recall	F1	mIoU	PA
Feature fusion	0.699	0.519	0.596	0.670	0.920
No feature fusion	0.683	0.443	0.533	0.638	0.914

4) Low-Level Feature Fusion Scheme of the Decoder: The proposed FFS-Net model modifies the decoder to enhance the reliability of landslide recognition by fusing both the low-level features yielded by the first block of the ResNet101 network and the high-level semantic information yielded by the SFE module, and its advantage is verified by the comparative experiments, whose results are listed in Table X. The experimental results show the low-level features contribute much to the pixel classification, and the mIoU metric has been improved by 3.2%.

VI. CONCLUSION

In this article, a multiple heterogeneous data FFS-Net was proposed for landslide detection, which can effectively extract morphological features from HRSIs and terrain features from DEM data, and then fuses these two source of information in a higher feature level. In the proposed FFS-Net model, a Siamese network without weight sharing was designed to separately extract these two types of features from the HRSIs and DEM data before fusing them. An improved Deeplabv3+ network is developed for better feature extraction, where multiscale channel attention module was developed in the encoder to automatically weight low-level fine features and high-level semantic features from ResNet101 to achieve an elegant balance between accurate detection and rich details, and transposed convolution layers replaced the original bilinear interpolation without learnable parameters in the decoder with result of better feature extraction capability. Moreover, dropout layers and BN layers were introduced in the decoder to prevent over-fitting and avoid gradient disappearance. The proposed model was evaluated on both Bijie and Loess Plateau datasets, and extensive comparison experiments were carried out to show that introducing DEM through feature fusion can present more valuable information and then greatly improve the detection accuracy of visually blurred old landslides and some new landslides that are visually similar to their surroundings. This work is the first on semantic segmentation for visually blurred old landslide detection through fusing heterogeneous HRSIs and DEM data.

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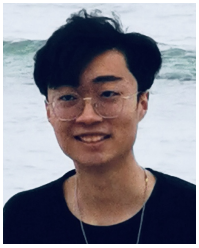
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